

# PLASMA SPRAY PROCESSORS

## Works Instruction

QF/QMS/02/08

Rev. 00

Date : 01/03/2019



Operations/Work/Job Activity covered by this assessment: Twin wire Arc Spray (TWAS)

### Required Personal Protective Equipment (PPE) :



**Safety glasses must be worn at times.**



**Long and loose hair must be contained.**



**Appropriate footwear with substantial uppers must be worn.**



**Close fitting/protective clothing to cover arms and legs must be worn.**



**Respiratory protection devices required.**



**Oil free leather gloves and spats must be worn.**



**Hearing protection must be used when using this machine.**

### Loading & Unloading Check:

1. Ensure no slip/trip is present in work pass and walk ways.
2. Ensure that work area is clean and clear of grease, oil and any flammable materials.
3. Refer Work Instructions PSP/WI/14 while handling.

### Operations Procedure:

1. Refer Pre-Inspection Work Instructions PSP/WI/12 to pre-inspect the Component.
2. Refer Final Blasting Work Instructions PSP/WI/04 for preparing surface. The surface preparation will be as per Sa3 specs.
3. Mask with High Temperature Adhesive Tape/Metal Shim/ to protect unwanted area from over spraying during TWAS spray coating.
4. Load the Component in the acoustic room and clamp it thoroughly on the chuck with proper packing to prevent clamping marks on the Component.
5. Start Exhaust Fans before starting the TWAS Control Unit.
6. Open valve of AIR and set the pressure according to the parameter mentioned in the Technical Bulletin.
7. Start the TWAS control unit with "On/OFF" switch.
8. Select the required Wire w.r.t the Job Card.
9. Measure the wire feed rate and Air pressure as per parameter, and set the Amperes and volts accordingly.
10. Set the job speed and Robot program speed, refer Job Card or instructions will be given by Supervisor/WM.
11. Closed the Acoustic Door and Wait for signal before igniting the Torch.
12. After ignition recheck the Amperes and Volt values and reset if required as per the desired parameter.

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13. Intermediate dimension should be checked to control dimension as per job card.C
14. Check the dimension to get the desired thickness of the coating mentioned in the job card.
15. After completion of spraying of the component
16. Cool the job gradually and recheck the dimension.
17. Fill up the Job Card as per the data and Tag it.

### Shutdown :

1. Shut the Switch and bring the Robot to its home position.
2. Cool the job gradually and recheck the dimension. Fill up the Job Card as per the data and Tag it.
3. Refer to Final Inspection Work Instruction PSP/WI/10 and get the approval Supervisor.

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|------------|--|------------|--|-------|--|
| Assessor : |  | Signature: |  | Date: |  |
| Approved:  |  | Signature: |  | Date: |  |